Experiment 1

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OBJECT ORIENTED PROGRAMMING

Q. Print Hello world.

A:

HelloWorld.java

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac HelloWorld.java } ; if ($?) { java HelloWorld }
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac HelloWorld.java } ; if ($?) { java HelloWorld }
Hello World
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. This experiment introduces students to **basic input and output operations** in Java. Students will write programs to accept data such as name, age, marks, etc. from the user using input statements and display the output using standard output statements.

A:

Student.java

```java
import java.util.Scanner;
public class Student {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        System.out.print("Enter your marks: ");
        float marks = sc.nextFloat();

        System.out.println("Name  : " + name);
        System.out.println("Age   : " + age);
        System.out.println("Marks : " + marks);

        sc.close();  // Close the Scanner
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac Student.java } ; if ($?) { java Student }
Enter your name: Archisha Borah
Enter your age: 20
Enter your marks: 83.4
Name  : Archisha Borah
Age   : 20
Marks : 83.4
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Printing one to N numbers.

A:

oneToN.java

```java
import java.util.Scanner;

public class oneToN {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter the number: ");
        int n = sc.nextInt();
        System.out.println("Numbers from 1 to " + n + " are as follows:");
        for (int i = 1; i <= n; i++) {
            System.out.println(i);
        }
        sc.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac oneToN.java } ; if ($?) { java oneToN }
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac oneToN.java } ; if ($?) { java oneToN }
Enter the number: 15
Numbers from 1 to 15 are as follows:
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Binary to Decimal.

A:

BinToDecimal.java

BinToDecimal.java > BinToDecimal > main(String[])

```java
import java.util.Scanner;
public class BinToDecimal {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println(x: "Enter the binary number: ");
        int binary = sc.nextInt();

        int ans = 0;
        int power = 0;

        while (binary > 0) {
            int digit = binary % 10;
            ans = ans + digit * (int) Math.pow(a: 2, power);
            power++;
            binary = binary / 10;
        }

        System.out.println("Decimal value: " + ans);
        sc.close();
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac BinToDecimal.java } ; if ($?) { java BinToDecimal }
Enter the binary number:
101
Decimal value: 5
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Decimal to Binary.

A:

DecToBinary.java

```java
import java.util.Scanner;
public class DecToBinary {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the number: ");
        int num = sc.nextInt();
        String binary = " ";
        while (num >0 ){
            int rem = num %2;
            binary= (rem + binary);
            num= num/2;
        }
        System.out.println(binary);
        sc.close();

    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac DecToBinary.java } ; if ($?) { java DecToBinary }
Enter the number:
22
10110
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Displaying employee details.

A:

employee.java > employee > main(String[])

```java
class employee {
    int ID;
    double salary;

    public void display() {
        System.out.println(ID + "\n" + salary);
    }

    Run | Debug
    public static void main(String[] args) {
        employee e1 = new employee();
        e1.ID = 22;
        e1.salary = 100000.22;
        e1.display();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac employee.java } ; if ($?) { java employee }
22
100000.22
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Find the largest number from three numbers.

A:

Largest3num.java

Largest3num.java > Largest3num > main(String[])

```java
import java.util.Scanner;
public class Largest3num {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter first number: ");
        int a = sc.nextInt();
        System.out.print(s: "Enter second number: ");
        int b = sc.nextInt();
        System.out.print(s: "Enter third number: ");
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number is: " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number is: " + b);
        } else {
            System.out.println("Largest number is: " + c);
        }
        sc.close();
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac Largest3num.java } ; if ($?) { java Largest3num }
Enter first number: 22
Enter second number: 19
Enter third number: 31
Largest number is: 31
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Sum of two numbers.

A:

user_input.java

user_input.java > user_input > main(String[])

```java

public class user_input {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number: ");
        int num1 = sc.nextInt();
        System.out.println(x: "Enter the second number: ");
        int num2 = sc.nextInt();
        int sum = num1+num2;
        System.out.println("The sum of two numbers is: " + sum);
        sc.close();
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac user_input.java } ; if ($?) { java user_input }
Enter the first number:
22
Enter the second number:
190
The sum of two numbers is: 212
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. LCM of two numbers.

A:

LCM.java

LCM.java > LCM > main(String[])

```java
import java.util.Scanner;
public class LCM {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number: ");
        int a = sc.nextInt();
        System.out.println(x: "Enter the second number: ");
        int b = sc.nextInt();
        int max = (a > b) ? a : b;
        while (true) {
            if (max % a == 0 && max % b == 0) {
                System.out.println("Lcm of " +a+ " and " +b+ " is " +max);
                break;
            }
            max++;
        }
        sc.close();
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac LCM.java } ; if ($?) { java LCM }
Enter the first number:
24
Enter the second number:
12
Lcm of 24 and 12 is 24
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Find the second largest element of an array.

A:

SecondLargestElement.java

SecondLargestElement.java > SecondLargestElement > main(String[])

```java
import java.util.Scanner;
public class SecondLargestElement {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter the number of elements in the array: ");
        int n = sc.nextInt();
        if (n < 2) {
            System.out.println(x: "Array must have at least two elements.");
            sc.close();
            return;
        }
        int[] arr = new int[n];
        System.out.println(x: "Enter the elements of the array:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int max1 = Integer.MIN_VALUE;
        int max2 = Integer.MIN_VALUE;
        for (int num : arr) {
            if (num > max1) {
                max2 = max1;
                max1 = num;
            } else if (num > max2 && num != max1) {
                max2 = num;
            }
        }

        if (max2 == Integer.MIN_VALUE) {
            System.out.println(x: "No second largest element found.");
        } else {
            System.out.println("The second largest element is: " + max2);
        }

        sc.close();
    }
}
```

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac SecondLargestElement.java } ; if ($?) { java SecondLargestElement }
Enter the number of elements in the array: 5
Enter the elements of the array:
Enter the number of elements in the array: 5
Enter the elements of the array:
22
13
56
78
13
56
78
56
78
78
37
The second largest element is: 56
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Find the third largest element of an array.

A:

ThirdLargest.java

ThirdLargest.java > ThirdLargest > main(String[])

```java
import java.util.Scanner;
public class ThirdLargest {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Ask for array size
        System.out.print(s: "Enter the number of elements in the array: ");
        int n = sc.nextInt();

        if (n < 3) {
            System.out.println(x: "Array must have at least three elements.");
            sc.close();
            return;
        }

        int[] arr = new int[n];

        // Take array input
        System.out.println(x: "Enter the elements of the array:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int first = Integer.MIN_VALUE;
        int second = Integer.MIN_VALUE;
        int third = Integer.MIN_VALUE;

        for (int i = 0; i < n; i++) {
            if (arr[i] > first) {
                third = second;
                second = first;
                first = arr[i];
            } else if (arr[i] > second && arr[i] != first) {
                third = second;
                second = arr[i];
            } else if (arr[i] > third && arr[i] != first && arr[i] != second) {
```

```java
                third = arr[i];
            }
        }

        if (third == Integer.MIN_VALUE) {
            System.out.println(x: "No third largest element found.");
        } else {
            System.out.println("The third largest element is: " + third);
        }

        sc.close();
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac ThirdLargest.java } ; if ($?) { java Third
Enter the number of elements in the array: 7
Enter the elements of the array:
23
4
58
26
900
366
13
The third largest element is: 58
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Check whether the number is odd or even.

A:

oddEven.java

oddEven.java > oddEven > main(String[])

```java
import java.util.Scanner;

public class oddEven {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter a number: ");
        int num = sc.nextInt();

        if (num % 2 == 0) {
            System.out.println(num + " is even.");
        } else {
            System.out.println(num + " is odd.");
        }

        sc.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac oddEven.java } ; if ($?) { java oddEven }
Enter a number: 27
27 is odd.
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac oddEven.java } ; if ($?) { java oddEven }
Enter a number: 22
22 is even.
PS D:\CSE\OBJECT ORIENTED PROGRAMMING>
```

Q. Simple calculator (2numbers).

A:

SimpleCal.java

SimpleCal.java > SimpleCal > main(String[])

```java
import java.util.Scanner;
public class SimpleCal {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print(s: "Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.println(x: "Choose an operation: ");
        System.out.println(x: "1. Addition (+)");
        System.out.println(x: "2. Subtraction (-)");
        System.out.println(x: "3. Multiplication (*)");
        System.out.println(x: "4. Division (/)");
        System.out.print(s: "Enter your choice (1-4): ");
        int choice = sc.nextInt();

        double result;

        switch (choice) {
            case 1:
                result = num1 + num2;
                System.out.println("Result: " + result);
                break;
            case 2:
                result = num1 - num2;
                System.out.println("Result: " + result);
                break;
            case 3:
                result = num1 * num2;
                System.out.println("Result: " + result);
                break;
            case 4:
                if (num2 != 0) {
                    result = num1 / num2;
                    System.out.println("Result: " + result);
                } else {
                    System.out.println(x: "Error: Division by zero is not allowed!");
                }
                break;
            default:
                System.out.println(x: "Invalid choice!");
        }

        sc.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac SimpleCal.java } ; if ($?) { java SimpleCal }
Enter first number: 22
Enter second number: 178
Choose an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 1
Result: 200.0
PS D:\CSE\OBJECT ORIENTED PROGRAMMING> cd "d:\CSE\OBJECT ORIENTED PROGRAMMING\" ; if ($?) { javac SimpleCal.java } ; if ($?) { java SimpleCal }
Enter first number: 22
Enter second number: 167
Choose an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 4
```